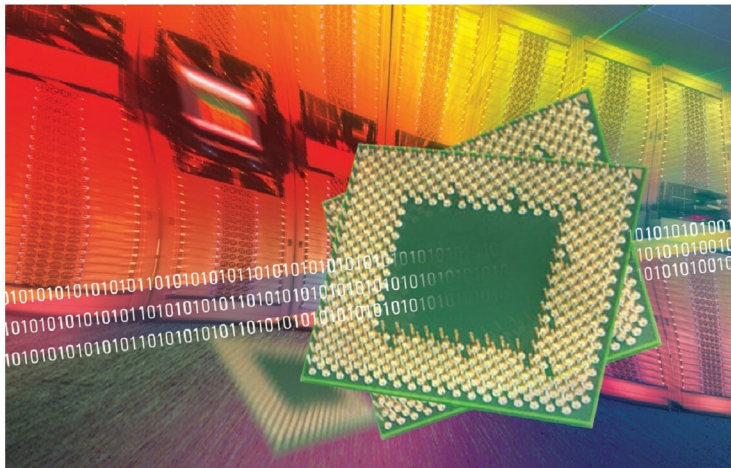


An Introduction to OpenCL

OpenCL (Open Computer Language) is a framework for writing programs that execute across heterogeneous platforms. It is based on the C99 language. Let's learn how to write a program using this framework.



OpenCL is a royalty-free, open standard for writing general-purpose parallel programs on heterogeneous computing platforms such as CPUs, GPUs, accelerators and FPGAs (field programmable gate arrays). OpenCL is based on the C99 language and a few added extensions to support data and task-based parallel programming models. The data-based parallel programming model or data parallelism is a form of parallel programming across multiple processors or computing elements, wherein each computing element executes the same piece of code on different data sets. Task-based parallel programs execute various lines of code on the same data set.

As mentioned earlier, in the OpenCL world, a host is connected with one or more computing units such as a CPU, GPUs (could be different vendors), accelerators or FPGAs. An OpenCL application is written as both host code and device code (called kernels). The device code (kernel) comprises C functions with some restrictions and a few extensions. The host code takes care of memory management, creating a work queue for each device, the data transfer

between host and device, and error management. The device code submitted to devices gets executed on them and these are generally compute-intensive in nature. In a nutshell, OpenCL abstracts the hardware details and hence the programmer need not focus on the device/vendor-specific code. The OpenCL platform model defines a device as an array of compute units, with each compute unit functionally independent from the rest. Compute units are further divided into processing elements.

As noted in Wikipedia, "OpenCL is an open standard maintained by the non-profit technology consortium, Khronos Group. It has been adopted by Apple, Intel, Qualcomm, Advanced Micro Devices (AMD), NVIDIA, Altera, Samsung, Vivante, Imagination Technologies and ARM Holdings."

The OpenCL Framework provides three main components, primarily the OpenCL runtime, APIs and the compiler. The runtime abstracts the underlying hardware from the operating system. The provided APIs help to create work queues (command queues), add kernel in the work queues and create context. The context in OpenCL